

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: CELL DIVISION (SUB-STRAND 1.4)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How does the body produce the enormous number of new cells needed for growth? Think back to the baby photographs and the time-lapse of a child growing over 14 years: a newborn of about 3 kg became a teenager of 50 kg or more. Use what you observed and learned in Lessons 1 to 10 to write your own Final Explanation. Answer each section in the blank space under the prompt, in full sentences. Use evidence from our investigations (diagrams, clay models, onion root tips, animations, case studies) and give reasons for your claims. You may add labelled sketches. Use correct scientific words such as chromosome, chromatid, daughter cell, mitosis, meiosis, zygote and gamete.

1. Claim: Where do the new cells come from?

A newborn of about 3 kg grows into a teenager of 50 kg or more, yet we only eat food such as ugali, sukuma wiki and beans. (a) Write a claim that answers the driving question. (b) Explain why eating food alone cannot account for the billions of new cells. (c) Compare this with your first ideas in Lesson 1 and say how your thinking has changed.

(a) The body makes the enormous number of new cells it needs for growth through cell division. Existing cells divide to form new cells, so the number of cells increases. (b) Food such as ugali, beans and sukuma wiki does not turn directly into cells. Food only gives the energy and raw materials, and cells use these to make new cells when an existing cell divides. The new cells always come from cells that already exist. (c) In Lesson 1 I thought cells might just grow bigger, multiply, or be made from food. Now I know that growth mainly happens because the number of cells increases by division, not only because cells become bigger.

2. Evidence: Mitosis and growth

Use evidence from the mitosis diagram, the clay model and the onion (kitunguu) root tip squashes. (a) Describe what you saw happen to the chromosomes as one cell became two, giving the stages of mitosis in the order they happen. (b) In the onion root tip,

(a) Mitosis has four stages. Chromosomes first became visible (prophase), then lined up in the middle of the cell (metaphase), then separated and moved to opposite ends (anaphase), and two new nuclei formed (telophase). Cytokinesis then followed, when the cytoplasm divided, giving two daughter cells with the same chromosomes as the parent cell. (b) Growing tissues such as the root tip have many cells in division, so new cells are being made all the time. Seeing cells at different stages shows that mitosis is continuous in places where growth happens. Cells spend most of their time in interphase between divisions. (c) Each round of mitosis doubles the number of cells: 1, 2, 4, 8, 16... If this repeats many times starting from the zygote, the number reaches billions. This is how a 3 kg baby can become a 50 kg teenager.

most cells were in interphase and a smaller number were dividing. What does this tell you about growing tissues? (c) Explain how a single fertilised cell could become billions of cells, using the pattern 1, 2, 4, 8, 16...	
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3. Reasoning: Why the new cells work properly

(a) What does a chromosome carry, and what is the structure of a chromosome before a cell divides? (b) Explain why DNA must be copied before a cell divides. (c) Explain how this makes sure each new body cell has the right instructions to work properly. (d) Name two body processes besides growth that need mitosis.	(a) A chromosome carries the genetic information (DNA and genes). Before a cell divides, the DNA is copied, so each chromosome is made of two identical chromatids joined at a centromere. (b) DNA is copied before division so that after the chromosomes are shared out, each daughter cell receives a full set of chromosomes. Without copying, the daughter cells would have too little genetic information. (c) In mitosis the copied chromosomes are lined up and then separated accurately, so one chromatid of each chromosome goes to each end of the cell. The two daughter cells therefore have the same chromosomes as the parent cell and the same instructions, so they can work properly as muscle, bone, skin or blood cells. (d) Repair of injured tissue, such as a healing skin cut, and replacement of cells. Mitosis also gives asexual reproduction in plants, for example new roots and shoots on a maize or sukuma wiki plant.
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4. Meiosis and reproduction

Every baby begins as one fertilised cell (zygote) formed when a sperm meets an egg. (a) If a sperm and an egg each had 46 chromosomes, how many would the zygote have, and what problem would that cause over generations? (b) Describe what meiosis does to the chromosome number and how many cells it makes. (c) Explain how meiosis and mitosis work together so that a child can be produced and then grow. (d) Give one difference between the cells made by mitosis and meiosis.	(a) The zygote would have $46 + 46 = 92$ chromosomes. The number would double in every generation, which cannot continue. (b) Meiosis happens in the reproductive organs and has two divisions (meiosis I and meiosis II). It produces four cells, each with half the chromosome number (23). In meiosis I homologous chromosomes separate and in meiosis II sister chromatids separate. (c) Meiosis makes the sperm and egg (gametes) with 23 chromosomes each. At fertilisation they fuse and the zygote has 46 chromosomes again. The zygote then divides by mitosis again and again to make the billions of cells of the growing child. (d) Mitosis makes two daughter cells that are genetically identical to the parent cell with the full chromosome number, while meiosis makes four cells with half the chromosome number, so it makes gametes rather than body cells.
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5. Other forms of cell division and their uses

(a) Describe how budding in yeast and binary fission in bacteria differ. (b) A single bacterium dividing every 20 minutes gives how many cells after 2 hours? Show how you worked it out, and say how this doubling is similar to what happens in mitosis in a growing child. (c) Explain how tissue culture of bananas or tea in Kenya uses the same idea as growth in a child. (d) Finish by writing a short Final Explanation (3 to 4 sentences) answering the driving question.	(a) In budding, a small bud grows on the parent yeast cell and then separates, so the parent and daughter cells are unequal. In binary fission, a bacterium copies its chromosome, elongates and splits into two equal cells. Both are asexual reproduction from one parent without fusion of sex cells, so offspring are identical to the parent. (b) Two hours is 120 minutes, which is six 20-minute divisions. Doubling six times gives $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$ cells. This is a calculation, not something we saw under the microscope, but it shows the same doubling pattern (1, 2, 4, 8, 16...) that we saw when mitosis makes two cells from one in the onion root tip and in a growing child. (c) In tissue culture a small piece of plant (explant) is placed in sterile nutrient medium. Its cells divide repeatedly by mitosis to form a callus, then shoots and roots, giving many identical disease-free plantlets. This is the same idea as a child's growth: existing cells divide by mitosis and each new cell gets a full copy of the chromosomes. (d) The body produces new cells through cell division. In mitosis one cell copies its DNA and divides into two identical daughter cells, so the cell number doubles again and again from the zygote, giving the billions of cells needed for growth, repair and replacement. Food such as ugali, beans and sukuma wiki only supplies energy and raw materials. Meiosis makes gametes with half the chromosome number for sexual reproduction, and yeast and bacteria reproduce by budding and binary fission. Therefore cell division is essential for the growth, development, repair and reproduction of living organisms.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that new cells come from division of existing cells, and clearly shows that food provides only energy and materials, not cells directly.	States that cell division makes new cells, but the role of food is unclear or only briefly mentioned.	Claim is missing, vague, or suggests food turns directly into cells or that cells only grow bigger.
Use of evidence from the investigations	Uses several specific observations (mitosis diagram, clay model, onion root tips, the 1, 2, 4, 8, 16 pattern) accurately to support the claims, and correctly treats the 64-bacteria figure as a calculation that shows the same doubling pattern rather than as an observation.	Uses one or two observations from the lessons, mostly accurate but general.	Gives little or no evidence, or evidence is inaccurate, not from the lessons, or calculations are presented as observations.
Scientific reasoning about chromosomes, mitosis and meiosis	Correctly explains DNA copying (chromatids identical after copying), the stages of mitosis in order, identical daughter cells, and how meiosis halves the chromosome number (23) so the zygote has 46; links both to the growing child.	Explains mitosis and meiosis mostly correctly but misses a link (for example the order of stages, or why DNA is copied or why the number is halved).	Confuses mitosis and meiosis or does not explain how new cells get the right instructions.
Breadth: growth, repair, reproduction and applications	Connects cell division to growth, repair, replacement, sexual and asexual reproduction (budding, binary fission) and	Covers growth and at least one other function or application accurately.	Mentions only growth, or applications are missing or incorrect.

	tissue culture with accurate Kenyan examples.		
Communication and use of scientific terms	Writes organised, complete sentences with correct terms (chromosome, chromatid, daughter cell, zygote, gamete) and a clear final explanation.	Mostly clear writing with some correct terms; final explanation is complete but may be loosely organised.	Writing is hard to follow, key terms are missing or misused, and the final explanation is incomplete.

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